Complex networks report: The Wester United States power grid

Adam César Maćkowiak Pellón

January 23, 2025

Abstract

This report analyzes the Western United States power grid through the framework of complex networks. Key structural properties, including density, clustering, path length, and centrality measures, are computed to characterize the network. The degree distribution is fitted to various statistical models, revealing that neither power-law nor Poisson distributions adequately describe the network. The network's characteristics are compared with theoretical models, including the Erdős–Rényi, Barabási–Albert, and Watts–Strogatz models, with none fully replicating the observed metrics. Community detection algorithms, such as the Louvain and greedy modularity optimization methods, identify a modular structure with over 40 communities and high modularity values. Visualizations of the network provide insights into its topology and community structure, highlighting nodes of high betweenness centrality that play critical roles in network resilience. This study underscores the importance of complex network theory in understanding power grid systems and their vulnerabilities.

1 Introduction

The power grid has been extensively analyzed through the lens of complex networks [1–16]. This perspective is highly beneficial as it allows the grid to be studied as an interconnected system. By understanding its structure and identifying vulnerabilities, we can propose strategies to strengthen it against failures, whether caused by deliberate attacks or natural events.

2 Methodology

This section outlines the tools and methodologies employed in this analysis.

2.1 Basic Metrics

The following metrics are used to characterize the network:

- **Density:** Measures the proportion of existing edges relative to the total possible edges in a network of n nodes and m edges:
$$\rho = \frac{2m}{n(n-1)}.$$

- **Clustering Coefficient:** Based on the Watts-Strogatz definition, the clustering coefficient for node i is:
$$C_i = \frac{2t_i}{k_i(k_i-1)},$$
where t_i is the number of triangles involving node i[1] and k_i is its degree. The average clustering coefficient across the network is:
$$\bar{C} = \frac{1}{n}\sum_{i=1}^{n} C_i.$$

- **Transitivity:** A global measure of clustering:
$$C = \frac{3|C_3|}{|P_2|},$$
where $|C_3|$ is the number of triangles and $|P_2|$ is the number of two-paths in the network.

[1] Also is the number of transitive relations.

- **Average Path Length:** The mean shortest distance between all pairs of nodes. Here $d(i,j)$ is the shortest distance between two nodes:

$$l_G = \frac{1}{n(n-1)} \sum_{i \neq j} d(i,j).$$

- **Diameter:** The longest shortest path between any two nodes in the network. Is the biggest value of $d(i,j)$ of the network.
- **Degree Assortativity:** Measures the tendency of nodes to connect with others of similar degree:

$$r = \frac{\sum_{(i,j)\in E} k_i k_j - \left(\sum_{(i,j)\in E} \frac{k_i+k_j}{2m}\right)^2}{\sum_{(i,j)\in E} \frac{k_i^2+k_j^2}{2m} - \left(\sum_{(i,j)\in E} \frac{k_i+k_j}{2m}\right)^2}.$$

- **Bipartivity:** The bipartivity index is used to quantify how close a network is to being bipartite. For a perfectly bipartite network, the index equals 1, while it decreases monotonically as the network becomes less bipartite. This index is computed as follows, where A represents the adjacency matrix of the network:

$$b_s = \frac{\mathrm{Tr}[\cosh(A)] - \mathrm{Tr}[\sinh(A)]}{\mathrm{Tr}[\cosh(A)] + \mathrm{Tr}[\sinh(A)]} = \frac{\mathrm{Tr}[\exp(-A)]}{\mathrm{Tr}[\exp(A)]}.$$

2.2 Centrality Measures

Several centrality measures are used to rank nodes:

- **Degree Centrality:** Ranks nodes by their degree. The nodes with the highest degree are the most central ones.
- **Closeness Centrality:** Normalized inverse of the total distance from a node to all other nodes:

$$CC(i) = \frac{n-1}{\sum_j d(i,j)}.$$

- **Betweenness Centrality:** quantifies the importance of a node by measuring how often it acts as a bridge along the shortest paths connecting pairs of other nodes. If $\rho(i,j)$ represents the total number of shortest paths between nodes i and j, and $\rho(i,k,j)$ is the number of those shortest paths that pass through node k, then the betweenness centrality of node k is defined as:

$$BC(k) = \sum_{i \neq j \neq k} \frac{\rho(i,k,j)}{\rho(i,j)}.$$

- **Eigenvector Centrality:** Based on the greatest eigenvalue λ_1 of the adjacency matrix, where $\vec{\varphi}_1$. If:

$$\vec{\varphi}_1(i) = \left(\frac{1}{\lambda_1} A \vec{\varphi}_1\right)_i.$$

- **Katz Centrality:** measures the influence of a node by considering the total weight of all paths that connect it to other nodes, with longer paths being exponentially attenuated. This metric depends on a free parameter α[2], which satisfies $0 < \alpha < \lambda_1^{-1}$, where λ_1 is the largest eigenvalue of the adjacency matrix A. If $\{\phi_q\}$ are the eigenvectors of the adjacency matrix, then the Katz centrality of node i is given by:

$$K(i) = \sum_q \sum_j \frac{\varphi_j(i)\varphi_j(q)}{1 - \alpha\lambda_j}.$$

[2] In Networkx, the default value of $\alpha = 0.1$.

- **PageRank Centrality:** Designed for directed graphs, evaluates the importance of a node based on its connectivity and the importance of its neighbors. For undirected graphs, the network is transformed to include bidirectional links for each edge. The matrix H is defined as:

$$H_{ij} = \begin{cases} \frac{1}{k_i^{\text{out}}} & \text{if there is a link from } i \text{ to } j, \\ 0 & \text{otherwise.} \end{cases}$$

This matrix is then converted into a row-stochastic matrix S as:

$$S = H + \vec{a}\left[\frac{1}{n}\vec{1}^T\right],$$

where $\vec{a}_i = 1$ if $k_i^{\text{out}} = 0$, and $\vec{a}_i = 0$ otherwise. Using a free parameter α[3], the final matrix is constructed as:

$$G = \alpha S + (1-\alpha)\frac{1}{n}\vec{1}\vec{1}^T.$$

If $\overrightarrow{PR}$ is the principal eigenvector of G corresponding to its largest eigenvalue, the PageRank centrality of node i is:

$$PR(i) = \left(G^T\overrightarrow{PR}\right)_i.$$

- **Subgraph Centrality:** Subgraph centrality evaluates the importance of a node based on its participation in all possible subgraphs of a network. For a network with adjacency matrix A, it is defined as:

$$EE(i) = [\exp(A)]_{ii}.$$

2.3 Statistical Distributions

Below are the Probability Density Functions (PDFs) for the different statistical fits. To fit them, Scipy has been used, where it uses the Maximum Likelihood Estimation method:

- **Power-law:** A power-law distribution, for $x > 0$, is described by the following expression:

$$f(x,\alpha) = Cx^{-\alpha}.$$

After using the Maximum Likelihood estimation, for $x \geq x_{\min}$, the PDF is given by:

$$f(x;\alpha) = \frac{\alpha-1}{x_{\min}}\left(\frac{x}{x_{\min}}\right)^{-\alpha}.$$

- **Exponential:** For $x \geq 0$, the PDF is expressed as:

$$f(x;\lambda) = Ce^{-\lambda x}.$$

- **Power-law + Exponential:** This is a combination of a power-law and an exponential cutoff, defined for $x \geq x_{\min}$ as:

$$f(x;\alpha,\lambda) = Cx^{-\alpha}e^{-\lambda x},$$

where C is the normalization constant.

- **Poisson:** For $k \in \{0,1,2,\ldots\}$, the PDF is:

$$f(k;\lambda) = \frac{\lambda^k e^{-\lambda}}{k!}.$$

[3] In Networkx, this time $\alpha = 0.85$ for this metric

- **Mielke:** The Mielke distribution for $x > 0$ is given by:

$$f(x, k, s) = \frac{kx^{k-1}}{(1 + x^s)^{1+k/s}}$$

2.4 Community Detection

We first look at the metrics used and then we will discuss the algorithms implemented in the code.

To assess the quality of the detected communities, several evaluation metrics are commonly used. These include modularity, performance, coverage, and conductance. Below, we describe these metrics in detail:

- **Modularity:** Modularity measures the density of edges within communities compared to the density of edges between communities. It evaluates how well the network is divided into communities. Formally, modularity is defined as:

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j),$$

  where A_{ij} is the adjacency matrix, k_i and k_j are the degrees of nodes i and j, m is the total number of edges, and $\delta(c_i, c_j)$ is 1 if nodes i and j belong to the same community, and 0 otherwise. A higher modularity value indicates a stronger community structure. After some algebra, one can reduce the expression from above to

$$Q = \sum_{c=1}^{n} \left[\frac{L_c}{m} - \gamma \left(\frac{k_c}{2m} \right)^2 \right],$$

  where the sum iterates over all communities c, L_c is the number of intra-community links for community c, k_c is the sum of degrees of the nodes in community c, and γ is the resolution parameter. The resolution parameter sets an arbitrary tradeoff between intra-group edges and inter-group edges. Nevertheless, it is common to just use $\gamma = 1$.

- **Performance:** Performance quantifies the proportion of correctly classified node pairs in the network. It considers both intra-community connections that exist and inter-community connections that do not exist. Mathematically, it is expressed as:

$$\text{Performance} = \frac{\text{Number of correct edges and non-edges}}{\text{Total number of possible edges and non-edges}}.$$

  This metric ranges from 0 to 1, where higher values signify better community separation.

  The way the algorithm determines if a pair of nodes is correctly classified is by looking at the structure of the graph and the structure determined by the community algorithm. If two nodes are connected by an edge and are considered part of the same community or if two nodes are not connected by an edge and are not part of the same community, then, in both cases, this pair of nodes is correctly classified.

- **Coverage:** Coverage evaluates the fraction of edges that are within the detected communities relative to the total number of edges in the network. It is calculated as:

$$\text{Coverage} = \frac{\text{Number of intra-community edges}}{\text{Total number of edges}}.$$

  While coverage provides insight into how well the algorithm groups nodes, it does not consider inter-community edges, so it is often used in combination with other metrics.

Below, we present the community detection methods employed:

- **Clauset-Newman-Moore Greedy Modularity Optimization**[17]: This approach seeks to maximize modularity. Initially, each node is placed in its own community. The algorithm iteratively merges nodes into communities that result in the greatest modularity increase until no further improvement is possible.

- **Louvain Method**[18]: This method is also based on modularity optimization and consists of two phases. In the first phase, each node is treated as its own community, and nodes are moved between communities to achieve maximum modularity. This phase concludes when no further gains can be made. In the second phase, a new network is constructed where nodes represent the communities from the first phase. Edges between nodes are replaced with weighted edges representing the sum of connections between communities. These steps are repeated until no further improvement in modularity is achieved.

2.5 Network Models

The Erdős-Rényi model[19] (Random Graph Model) generates a random graph where each pair of N nodes has an independent probability p of being connected by an edge. This model creates a network with a binomial degree distribution, which approaches a Poisson distribution for large N. While it is useful for modeling random processes, it lacks clustering and does not reproduce the scale-free properties observed in many real-world networks. This model is often applied to study randomly connected systems, such as communication networks or certain biological networks. Due to their construction method, Erdős-Rényi networks exhibits a binomial distribution that, for large values of n, it approximates to a Poisson distribution

$$p(k) = \binom{n-1}{k} p^k (1-p)^{n-1-k} \xrightarrow{n\to\infty} p(k) = \frac{\bar{k}^k}{k!} \exp(-\bar{k}),$$

where $\bar{k}$ is the average degree of the network.

On the other hand, The Barabási-Albert model[20] (scale-free model) generates networks using the concept of *preferential attachment*, where new nodes are more likely to connect to existing nodes with higher degrees. This process produces *scale-free* networks characterized by a power-law degree distribution, which captures real-world phenomena such as the "rich-get-richer" effect seen in social networks, the internet, and citation networks. This model is widely used to study systems that exhibit hierarchical and hub-based structures. This model generates networks with a degree distribution that adheres to a power-law distribution:

$$p(k) = \frac{2d(d-1)}{k(k+1)(k+2)} \xrightarrow{k\to\infty} k^{-3}.$$

Finally, The Watts-Strogatz model[21] (small-world model), which is not mandatory for this report but has been implemented just for the sake of completeness, begins with a regular lattice and rewires edges with a probability p, resulting in a network that combines local clustering with global shortcuts. This model produces networks that exhibit the *small-world* property, characterized by high clustering coefficients and short average path lengths. By varying the rewiring probability p, the model transitions from a regular lattice ($p = 0$) to a random graph ($p = 1$). It is used to study networks such as social interactions, neuronal networks, and collaboration structures. The degree distribution for the Wats-Strogatz model is not well defined.

2.6 Network specifics

The network under study corresponds to the Western US power grid network, where each node represents transforms or power relay points and two nodes are connected by an edge if power line runs between them. The network has been download from *Netzschleuder* and is undirected and unweighted. All nodes have at least one connection to another node.

3 Results

3.1 Generalm measurments of the network

To begin, let us summarize some of the key structural properties of the network under investigation.

The network has a total of 4941 nodes and 6594 edges, with an average degree of $\langle k \rangle = 2.67$ and a standard deviation of $\sigma_k = 1.79$. Finally, when we compute the density, we obtain $\rho = 5.40 \cdot 10^{-4}$ These

results align with values reported in the literature [1]. This outcome is expected since each edge represents a power line, which incurs costs and maintenance, thus limiting the network's overall density.

On the other hand, the average clustering coefficient is $\overline{C} = 0.0801$ and the transitivity index is $C = 0.103$ while the average path length is $\ell_G = 18.99$, and the network's diameter is $D = 46$. We see that, indeed, the network is large and the clustering is similar both globally and locally.

The assortativity coefficient of the network is $r = 0.00346$, a value close to zero. This indicates that the nodes do not preferentially connect based on degree, showing no clear tendency for nodes to attach to others with similar or dissimilar degrees.

Additionally, the bipartivity index was calculated as $b_s = 0.726$. This shows that the network is far from being bipartite in structure.

Finally, we will show the top 25 most central nodes according to different centrality measures in the following table 1. We also have ploted a matrix to compare the number of common nodes between each type of centrality measurement.

Rank	Degree	Closeness	Betweenness	Eigenvector	Katz	PageRank	Subgraph
1	2553	1308	4164	4381	4345	4458	4345
2	4458	2594	2543	4345	4381	831	4381
3	831	2605	1243	4336	4336	3468	4352
4	3468	1131	4219	4332	4352	2553	4384
5	4345	2606	2528	4352	4332	1224	4336
6	2382	1243	1267	4384	4395	597	4395
7	2542	1476	1308	4402	4384	2382	4332
8	2575	2557	1244	4347	4373	2575	4373
9	2585	2528	426	4395	4402	2439	4402
10	3895	2532	2606	4373	2553	3895	4361
11	1224	2612	2594	4361	4361	3355	4401
12	2434	2543	2605	4398	4347	2434	4347
13	2439	4219	69	4401	4392	2542	4392
14	2617	1125	108	4413	4401	2617	4413
15	2662	2618	1167	4408	4372	2647	2553
16	1005	4164	4120	4335	4346	2282	4408
17	1309	1267	2235	4392	4413	1005	4398
18	1334	1383	70	4344	4458	4520	4344
19	2282	427	2223	4346	4398	1030	4372
20	490	426	393	4372	4342	1334	4374
21	4332	1307	1476	4342	4391	2585	4335
22	4352	2909	1125	4374	4408	1326	4342
23	4373	2486	427	4391	2542	4878	4346
24	4381	4120	1131	4365	4374	490	4376
25	4384	2953	2557	4376	4335	4224	2662

Table 1: Ranking of the top 25 most central nodes following different types of centrality measures.

By analyzing the rankings presented in table 1 and the figure 1, we can observe that the closeness and betweenness centrality rankings have a large quantity of nodes in common, 17 to be precise, while they do not share any node with any other type of centrality measurement. On the other hand, Eigenvector centrality, Katz centrality and subgraph centrality share almost every single node in their lists. Finally, the degree centrality and the PageRank centrality also share 17 nodes and, again PageRank centrality does not have in common any or almost any node with the rest of the metrics, but the degree centrality, unlike the previous one, does share nodes with the rest of measurements, except, again, with closeness centrality and betweenness centrality.

This distinction is logical because closeness and betweenness centralities are more dependent on edge-related metrics and the distances between nodes in the network. Conversely, the other centrality measures are tied to node-related properties and their degree. Within this second group, the eigenvector centrality, Katz centrality, and subgraph centrality produce rankings that are highly consistent with each other.

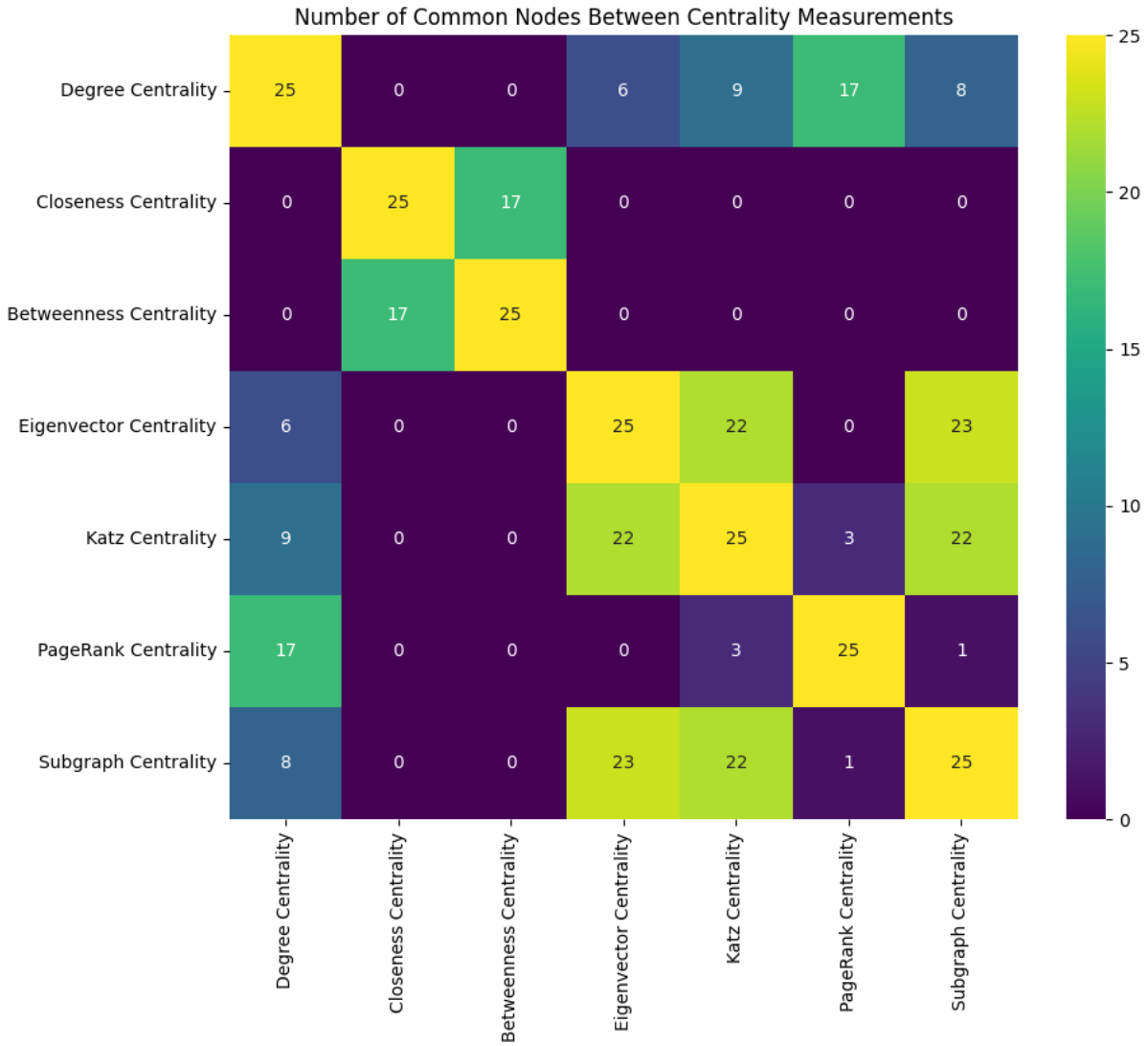

Figure 1: Matrix that plots the number of common nodes between each type of centrality measurement.

Similarly, the PageRank centrality aligns closely with the degree centrality, which is expected given their shared dependence on node degree.

The fact that closeness and betweenness centralities yield rankings that differ significantly from the other centrality metrics can be attributed to the nature of the power grid network. Such networks typically consist of small, well-connected clusters within specific regions, interspersed with nodes that act as connectors between clusters. When rankings are based on degree or similar metrics, the most central nodes tend to be the "core" of each cluster. In contrast, rankings derived from closeness and betweenness centralities prioritize the nodes that serve as connectors between clusters.

Then, is logical to think that one of the most relevant types of centralities in the study of power grids are betweenness centralities and information centralities[11], being equivalent in some cases. We will focus on the already studied betweenness centrality. This metric plays a very important role on analyzing cascades attacks and vulnerabilities of the network, a hot topic on the study of power grids [2, 3, 9–11, 13–16].

3.2 Degree distributions and fits

We have fitted the degree distribution to a different number of PDF, as you see in 2. The parameters obtain for each fit are displayed in table 2. To do it, we have fitted PDF to different functions and then computed the CDF and the complementary cumulative distribution function (CCDF), which let us see better how well is the tail fitted, usually the point of biggest interest. In all the cases we have used *scipy*,

where it uses the likelihood estimation method and in all cases the first started in the first point of the degree distribution for the data collected. That is why almost all start around the first point of the degree distribution and because of that no fit starts around 0 in the CDF and the CCDF but around the first point of the data. Neverthess, in both cases the distribution and the fits have been normalized such that they end in one and zero respectively. For the power-law fit, the module *powerlaw* has also been used (is the fit called "power-law fit 2"). Surprisingly it fits differently but even more surprisingly is the fact that the value of the $xmin$ was a decimal number less than one. And if we look at the other power-law fit, we see that the error estimation is bigger than the value of the constant. My interpretation is that the power-law is not a good for this network and it can already be seen in the graph neither one performs very well. It makes sense since, due to economical constraints, it cannot exhibit a heavy tail.

Looking at the tails, we can also see that neither the "Power-law + Exp fit" nor the "Poisson fit" do very well. In the last case it makes sense, since the powergrid is a well stablished network, done will careful thought, and it is not random at all. Another very important constraint which is relevant for all fits is the fact that the power grid is a planar graph. Because of that, is logical that the Poisson distribution does not fit properly. Nevertheless, is surprising that even this fit is better than the power-law fit. Also, λ should be equal to $\langle k \rangle$ and we see that this is not the case.

The best fits for the tail by far are the Mielke and the exponential fits, which is also reflected when we look at the complementary cumulative distribution function. The last case is not a surprise since there have been a lot of studies that use the exponential distribution for the study of powergrid networks. An extensive research with different fits has been done already and can be found in the literature [16] and how different fits are better for different levels of voltages.

Last, but not least, the CDF let us see better how well does the fit at the beginning. We see that, although both Poisson and Power-law + Exponential distribution do not fit greatly for the tail, they do in fact fit very nicely at the beginning, even better than the Exponential fit. This will be reflected once we look at the log-likelihood for each fit.

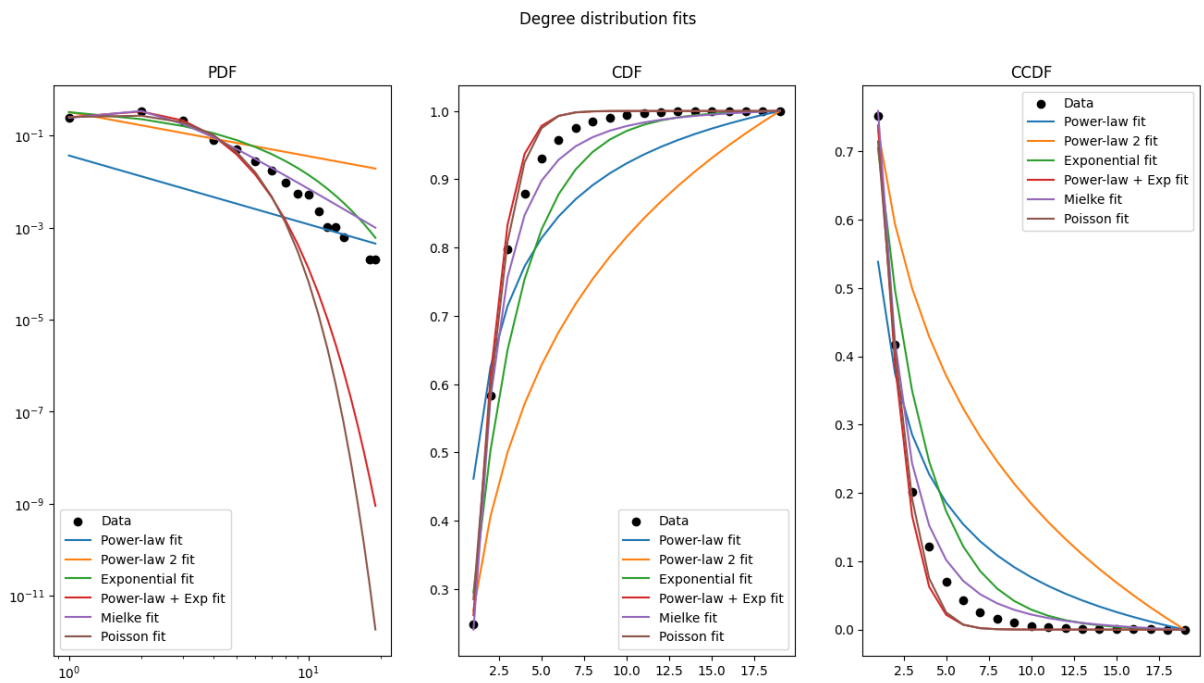

Figure 2: PDF, CDF and CCDF of the real network fitted by multiple models.

We see that the first power-law has the lowest log-likelihood by a very large margin, but then surprisingly, exponential does worst than the Poisson fit, the Mielke fit and the Power-law + Exponential fit. This is because it considers every part of the distribution equally important. This means that, the beginning of the distribution is as important as the tail, which is not true[4].We also have to remember that the scale for the PDF is logarithmic, so we have to be careful to not have a wrong impression about the distances.

[4]Again, one is usually more interested in the behavior at the tail of the distribution than at its beginning.

Distribution	Parameters	log-likelihood
Power-law	$\alpha = 1.5 \pm 0.4$, $x_{min} = 0.005 \pm 0.000$	-2299.34
Power-law 2	$C = 0.3 \pm 0.4$, $\alpha = 0.96 \pm 0.0$	28.47
Exponential	$C = 0.46 \pm 0.07$, $\lambda = 0.35 \pm 0.06$	36.44
Power-law + Exp.	$C = 1.12 \pm 0.08$, $\alpha = -2.59 \pm 0.15$, $\lambda = 1.50 \pm 0.08$	67.19
Mielke	$k = 8.6 \pm 0.4$, $s = 2.08 \pm 0.05$	62.11
Poisson	$\lambda = 2.14 \pm 0.09$	49.89

Table 2: Table with the parameters computed for all the fits of all the fits

Balancing both the importance of the tail and the statistical test, the Mielke fit seems to be the best option.

Lastly, the log-likelyhood serves to compare between different fits, but does not take into account the fact that the Power-law + Exponenital fit does use three or the Poisson uses only one. To have that into consideration, one should use the Akaike Information Criteria (AIK) [22]. In this report we have not implemented it because the fits did not have a lot of parameters.

3.3 Model comparisons

In this part of the report we will compare the real network with the Erdös-Rényi model, the Barabási-Albert model and the Watts-Strogatz model. To do that, we have run ten simulations with each model and have measured the same metrics we have already measured in the real network. Just for the sake of completeness, I will also compare the PDF and the CDF of all the models[5].

3.3.1 Erdös-Rényi model

There are different ways to simulate a Erdös-Rényi graph. The one that has been chosen for this project is $nx.gnm_random_graph(n, m)$, where a graph is chosen uniformly at random from the set of all graphs with n nodes and m edges. One important note is that these networks will be highly disconnected. Our solution was to choose in each realization the largest connected component, but keep in mind that the statistics will not be as good as one might expect. We obtained the following degree distribution:

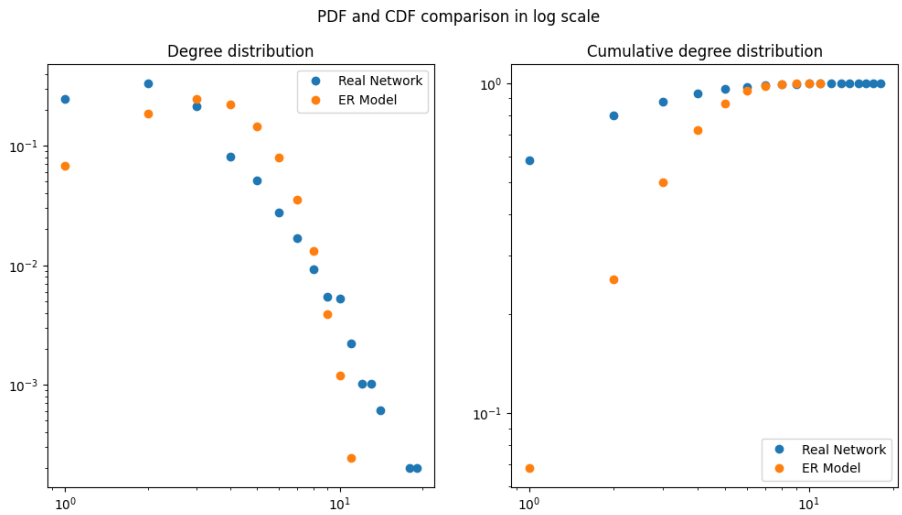

Figure 3: PDF and CDF of both the real network and the average of the 10 simulations of the Erdös-Rényi model.

From the ER model we could expect to have something similar to a Poisson Distribution, but again, the statistics are not the best. It has some resemblance with the Poisson fit shown in 2 when we look at the PDF, especially at the tail, but then we compare it to the CDF fit, we see that are not as similar

[5]The CCDF was not computed this time because, as it is shown, the comparisons between our network and these models are not very good and the CCDF would not bring anything new to the discussion.

and have noticeable discrepancies. Also, if we look at the point on the graph, we can cleary see that the graphs simulated are highly disconnected, because we only have data for low values of k.

3.3.2 Barabási-Albert

For the Barabási-Albert model we have $nx.barabasi_alber_graph(n, m)$, where n is the number of nodes the graph will have and m is the number of edges to attach from a new node to existing nodes. Here n is the number of nodes of the real network, while we used $m = 2$. Higher values of m will give place to more connected networks, and lower values of this quantity implies that the nodes with highest degree monopolize even more the connections, which results on a heavier tail. The truth is that the tail of our real world network does not seem to be very heavy so maybe higher values of m give raise to a degree distirbution more similar to the one from the power grid.

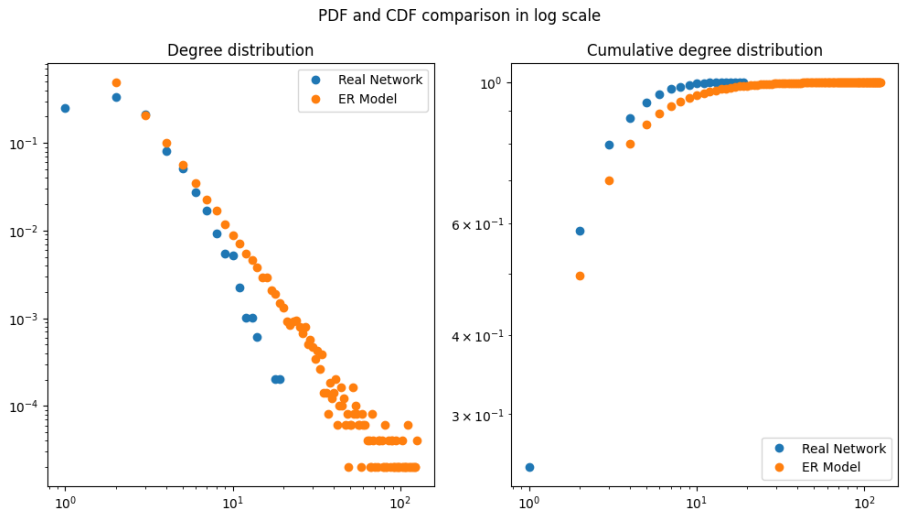

Figure 4: PDF and CDF of both the real network and the average of the 10 simulations of the Barabási-Albert model.

The Barabási-Albert model follows a power-law distribution for high values of k. Nevertheless, we can see the highest degree node from the simulation has a $k \sim 100$, so maybe could work better for more networks with more nodes and more connections. We see that both the pdf and cdf do not fit very well with the actual degree distribution of our network, as we have already see in the power-law fit before 2.

Also, we see that there is no node with $k = 1$, which makes sense since $m = 2$.

3.3.3 Watts-Strogatz

For the Watts-Strogatz model we used $watts_strogatz_graph(n, k, p)$, where it creates a network with a ring topology with n nodes, all connected to $k = 4$ neighbours initially and with a rewiring probability $p = 0.2$. If $k = 3$ the function will connect all the nodes initially to only two other nodes, so either we use $k = 2$ or $k = 4$ if we want to make a network with k as similar as possible to our $\langle k \rangle$. As we can see it does not fit very well with the degree distribution. We also see that the simulations do not have nodes with $k = 1$, which makes sense because the rewiring probability is relatively low. This also would explain why the maximum maximum degree nodes has just $k = 7$. Maybe different values of p and k could make the PDF and CDF more similar to those from our power grid.

3.3.4 Comparison between all the measurements

In the following table we show the metrics obtained for the different models and we compare them to the ones from our real world network.

We see that there is no metric that is similar to the ones from the real world network, which is quite interesting. Again, maybe playing a bit with the parameters one can obtain metrics more similar to the ones from our power grid.

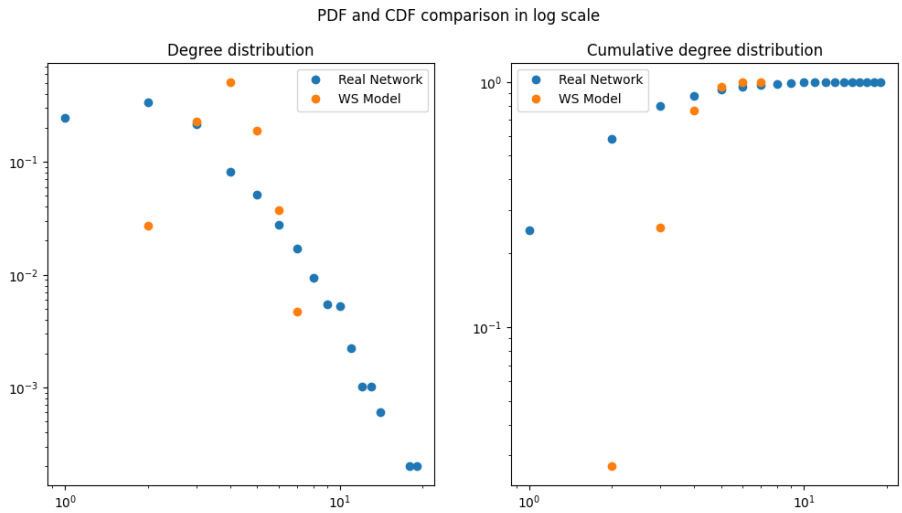

Figure 5: PDF and CDF of both the real network and the average of the 10 simulations of the Watts-Strogatz model.

Metric	Real Network	Erdös-Rényi	Barabási-Albert	Watts-Strogatz
$\bar{C}$	0.080104	0.0005 ± 0.0003	0.0079 ± 0.0007	0.263 ± 0.005
C	0.103153	0.0006 ± 0.0003	0.0028 ± 0.0003	0.239 ± 0.006
l_G	18.989185	8.49 ± 0.04	4.73 ± 0.05	8.77 ± 0.07
$max\{d(i,j)\}$	46.000000	19.30 ± 1.19	8.30 ± 0.46	15.9 ± 0.7
r	0.003457	0.001 ± 0.011	-0.055 ± 0.005	-0.048± 0.009
b_s	0.725917	0.9988 ± 0.0004	0.41 ± 0.06	0.629 ± 0.006

Table 3: Metrics measured for the different models and compared with the ones obtained in the real network.

The clustering coefficient $\bar{C}$ of the real network is 0.080104, indicating a relatively low level of clustering. The Erdős-Rényi model ($\bar{C} = 0.0005 \pm 0.0003$) and the Barabási-Albert model ($\bar{C} = 0.0079 \pm 0.0007$) significantly underestimate clustering compared to the real network. In contrast, the Watts-Strogatz model produces a clustering coefficient of 0.263 ± 0.005, which overestimates it.

The transitivity C of the real network is 0.103155, representing a slightly higher global clustering coefficient than the local clustering. The Erdős-Rényi model ($C = 0.0006 \pm 0.0003$) and the Barabási-Albert model ($C = 0.0028 \pm 0.0003$) ,again, drastically underestimate this value and, again, the Watts-Strogatz model ($C = 0.239 \pm 0.006$) overestimates transitivity.

The average shortest path length l_G for the real network is 18.989185, suggesting a less efficient structure for information transfer. The Erdős-Rényi model ($l_G = 8.49 \pm 0.04$) and the Barabási-Albert model ($l_G = 4.73 \pm 0.05$) both produce significantly shorter path lengths, underestimating the real-world value. The Watts-Strogatz model, with $l_G = 8.77 \pm 0.07$, provides a better approximation but still falls short of the real-world network's characteristics.

The diameter $\max\{d(i,j)\}$ of the real network is 46.000000, indicating a large maximum shortest path. The Erdős-Rényi model (19.30 ± 1.19) and the Watts-Strogatz model (15.9 ± 0.7) significantly underestimate the diameter. The Barabási-Albert model (8.30 ± 0.46) produces an even lower value, suggesting that it fails to replicate the large-scale connectivity observed in the real network.

The assortativity coefficient r for the real network is 0.003457, indicating a weak correlation between the degrees of connected nodes. The Erdős-Rényi model (0.001 ± 0.011), which has great error estimations than the estimated value, and the Barabási-Albert model (-0.055 ± 0.005) deviate from this value, with the latter showing slight disassortative mixing. The Watts-Strogatz model (-0.048 ± 0.009) also displays disassortative behavior, though it is slightly closer to the real network. In this aspect, maybe the model that is more similar to the power grid is the Erdös-Rényi, which gives an assortivity close to zero.

The bipartivity coefficient b_s for the real network is 0.725917, reflecting its structural balance. The Erdős-Rényi model (0.9988 ± 0.0004) greatly overestimates this metric, while the Barabási-Albert model (0.41 ± 0.06) underestimates it. The Watts-Strogatz model (0.629 ± 0.006) provides a closer approximation but still falls short of the real network value.

All of this results make sense. A power grid is a peculiar network because is a planar graph, which is a very heavy constraint, and does not follow the typical conventions of social networks. It also has very important economic constraints. All of this makes the system to not have heavy tails and is also not random. Because of that, no model captures correctly the properties of the network. An alternative that could work is to use a model that generates a planar network, like a random geometric graph. But this is something that can be explored in future work.

3.4 Communities detection

For the community detections we used two algorithms: The Louvain algorithm and the greedy modularity algorithm. On the other hand we used different metrics like the modularity, the performance and the coverage, which are all explained in section 2.4. We also measured the number N_C of communities obtained. One brief comment about the Louvain: this algorithm depends on the order at which the nodes are considered, so, considering this selection is random, we run the code 100 times to do an estimation of the value, where the error estimation is computed as

$$\delta_x = \sigma_x / \sqrt{M},$$

where $M = 100$ is the number of simulations we did. Having said this, here are the results.

Algorithm	N_C	Modularity	Performance	Coverage
Louvain	40.34 ± 0.18	0.93578 ± 0.00006	0.96571 ± 0.00012	0.97125 ± 0.00011
Greedy optimization	43	0.93457	0.96709	0.96870

Table 4: Metrics measured for the different models and compared with the ones obtained in the real network.

As we can see, the number of communities found by the Louvain algorithm is higher than those found by the Greedy Modularity Optimization algorithm, as well as the modularity and Coverage. On the other hand, performance seems to be higher in the second algorithm than in the first one. Nevertheless, the differences on the results obtained by both methods are minimal. In both cases, all the metrics are near one. Modularity is a measure of the strength of division of a network into communities. A high modularity value indicates that the network exhibits a well-defined community structure, where nodes within the same community are densely connected, while connections between nodes in different communities are relatively sparse. On the other hand, high performance indicates that the partition maximizes the number of edges within the communities and minimizes the number of edges between communities. And finally, high conductance value indicates that the communities are internally cohesive, with strong internal connectivity and weak external connections. These results will be very clear in the following section, where we show two visualizations of the network.

3.5 Visualization

Finally, we did three visualizations of the power grid. The first one 6 uses the positions obtained from the *nodes.csv*, probably representing the geographical positions of the transforms or power relay points. The other two will be displayed in the appendix. For the communities, the Louvain algorithm and each one was painted with a different color. We highlighted the node with the highest betweenness centrality in each community. If these nodes were attacked or failed, the overall efficiency of the network could decrease significantly, potentially triggering a cascade failure across the network.

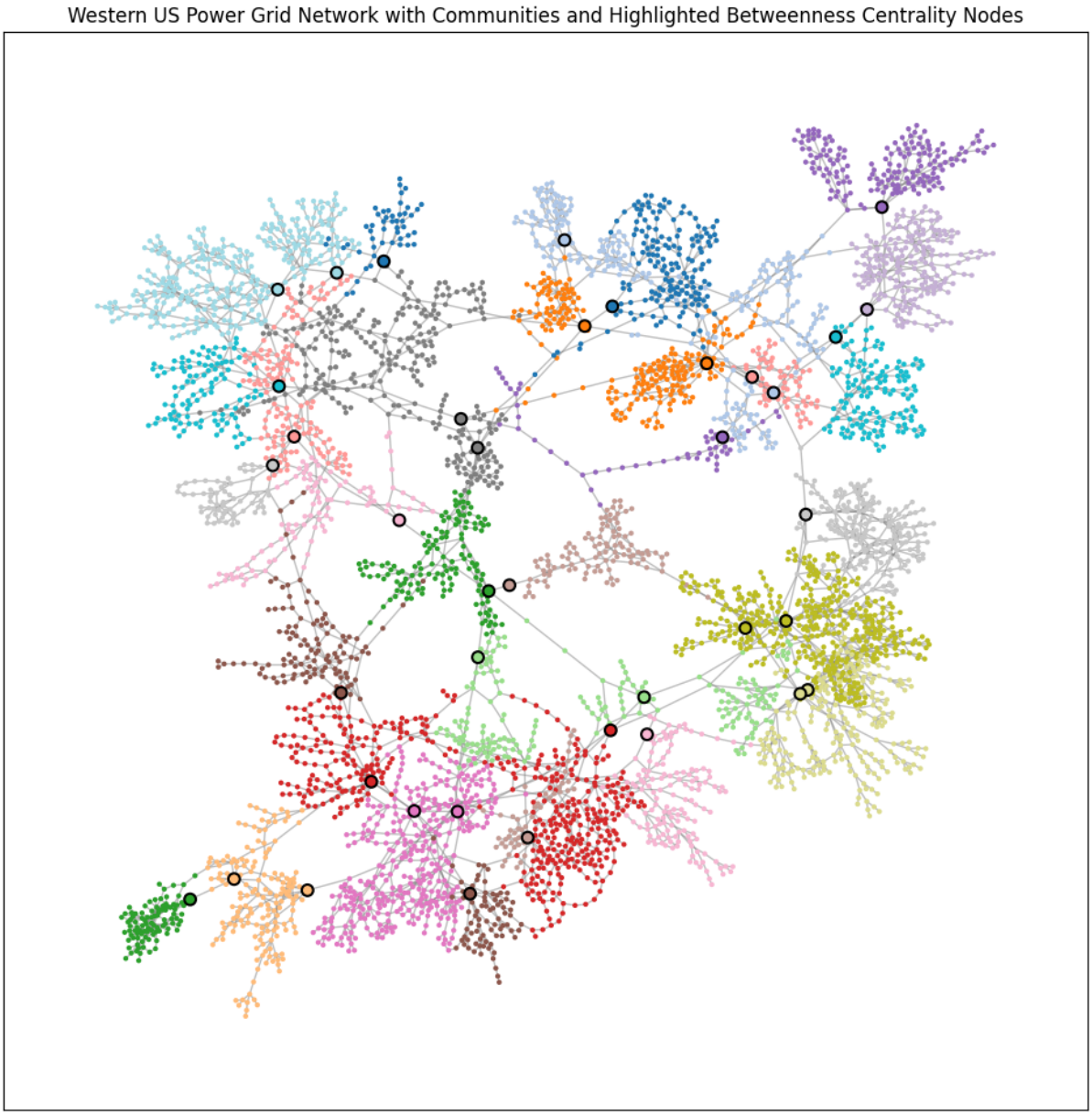

Figure 6: Visualization of the network with the real world layout.

4 Conclusions

This study provides an extensive analysis of the Western United States power grid using the tools of complex network theory. The network exhibits unique structural properties, including low density, low clustering coefficients, and a long average path length, reflecting its design constraints and operational needs. Centrality measures reveal the critical nodes that are likely to influence the network's resilience and efficiency, particularly in the context of cascading failures or targeted attacks, highlighting nodes with the highest betweenness centrality because they represent the vulnerabilities of the network.

The degree distribution analysis shows that the network does not follow standard statistical models such as power-law or Poisson distributions, due to economic constraints and the fact that the power grid is a planar network. Comparisons with theoretical network models highlight significant differences in all departments. Watts-Strogatz model caught some clustering properties but nothing remarkable, and the same could also be said with the assortativity obtained by the Erdös-Rényi model.

Community detection algorithms identify a modular structure with high modularity and performance, emphasizing the presence of well-defined communities. These communities align with the network's

functional substructures, and their visualization provides further insight into the distribution of critical nodes.

Overall, this report demonstrates the utility of complex network analysis in understanding power grid systems, particularly for identifying vulnerabilities and optimizing design.

References

[1] Netzschleuder. *Power Network Dataset*. Accessed: 2025-01-08. 2023. URL: https://networks.skewed.de/net/power.

[2] Yali Zhu, Nengfei Zhu, Gang Chen, and Weihua Gu. "Centrality Measures for Power Grid Network Based on K-shell Method". In: *2021 3rd International Academic Exchange Conference on Science and Technology Innovation (IAECST)*. 2021, pp. 1664–1669. DOI: 10.1109/IAECST54258.2021.9695879.

[3] Zhifang Wang, Anna Scaglione, and Robert J. Thomas. "Electrical centrality measures for electric power grid vulnerability analysis". In: *49th IEEE Conference on Decision and Control (CDC)*. 2010, pp. 5792–5797. DOI: 10.1109/CDC.2010.5717964.

[4] P. Hines, S. Blumsack, E. Cotilla Sanchez, and C. Barrows. "The Topological and Electrical Structure of Power Grids". In: *2010 43rd Hawaii International Conference on System Sciences*. 2010, pp. 1–10. DOI: 10.1109/HICSS.2010.398.

[5] Zhifang Wang, Anna Scaglione, and Robert J. Thomas. "Generating Statistically Correct Random Topologies for Testing Smart Grid Communication and Control Networks". In: *IEEE Transactions on Smart Grid* 1.1 (2010), pp. 28–39. DOI: 10.1109/TSG.2010.2044814.

[6] Zhifang Wang, Anna Scaglione, and Robert J. Thomas. "Generating Statistically Correct Random Topologies for Testing Smart Grid Communication and Control Networks". In: *IEEE Transactions on Smart Grid* 1.1 (2010), pp. 28–39. DOI: 10.1109/TSG.2010.2044814.

[7] Brian Cloteaux. "Limits in Modeling Power Grid Topology". In: *Proceedings of the IEEE International Conference on Infrastructure Systems and Services*. 2013, pp. 1–8.

[8] Paul Schultz, Jobst Heitzig, and Jürgen Kurths. "A Random Growth Model for Power Grids and Other Spatially Embedded Infrastructure Networks". In: *The European Physical Journal Special Topics* 223.12 (2014), pp. 2593–2610. DOI: 10.1140/epjst/e2014-02279-6.

[9] Deepjyoti Deka and Sriram Vishwanath. "Analytical Models for Power Networks: The Case of the Western US and ERCOT Grids". In: *IEEE Transactions on Power Systems* 30.2 (2015), pp. 1–10.

[10] Jian-Wei Wang and Li-Li Rong. "Cascade-Based Attack Vulnerability on the US Power Grid". In: *Safety Science* 47.10 (2009), pp. 1332–1336. DOI: 10.1016/j.ssci.2009.02.002.

[11] Paul Hines and Seth Blumsack. "A Centrality Measure for Electrical Networks". In: *Carnegie Mellon Electricity Industry Center Working Paper* CEIC-07 (2007). URL: www.cmu.edu/electricity.

[12] Siddharth Bhela, Harsha Nagarajan, Deepjyoti Deka, and Vassilis Kekatos. "Efficient Topology Design Algorithms for Power Grid Stability". In: *arXiv preprint arXiv:2103.05194* (2021). URL: https://arxiv.org/abs/2103.05194.

[13] Jian Zhou, Stamatis Tsianikas, David W. Coit, and Frank A. Felder. "Resilience based optimization for western U.S. transmission grid against cascading failures". In: *Proceedings of the 2019 IISE Annual Conference*. 2019.

[14] Jian-Wei Wang and Li-Li Rong. "Robustness of the western United States power grid under edge attack strategies due to cascading failures". In: *Safety Science* 49.6 (2011), pp. 807–812. DOI: 10.1016/j.ssci.2010.10.003.

[15] Samuel Forsberg, Karin Thomas, and Mikael Bergkvist. "Power grid vulnerability analysis using complex network theory: A topological study of the Nordic transmission grid". In: *Physica A: Statistical Mechanics and its Applications* 626 (2023), p. 129072. DOI: 10.1016/j.physa.2023.129072.

[16] Giuliano Andrea Pagani and Marco Aiello. "The power grid as a complex network: A survey". In: *Physica A: Statistical Mechanics and its Applications* 392 (2013), pp. 2688–2700. DOI: 10.1016/j.physa.2013.01.023.

[17] Aaron Clauset, M. E. J. Newman, and Cristopher Moore. "Finding community structure in very large networks". In: *Physical Review E* 70.6 (2004), p. 066111. DOI: 10.1103/PhysRevE.70.066111.

[18] Vincent D Blondel, Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefebvre. "Fast unfolding of communities in large networks". In: *Journal of statistical mechanics: theory and experiment* 2008.10 (2008), P10008.

[19] P. Erdős and A. Rényi. "On the evolution of random graphs". In: *Publicationes Mathematicae* 6 (1959). Accessed: 2025-01-08, pp. 290–297. URL: https://www.renyi.hu/~p_erdos/1959-11.pdf.

[20] Albert-László Barabási and Réka Albert. "Emergence of Scaling in Random Networks". In: *Science* 286.5439 (1999), pp. 509–512. DOI: 10.1126/science.286.5439.509. URL: https://science.sciencemag.org/content/286/5439/509.

[21] Duncan J Watts and Steven H Strogatz. "Collective dynamics of 'small-world'networks". In: *nature* 393.6684 (1998), pp. 440–442.

[22] H. Akaike. "A new look at the statistical model identification". In: *IEEE Transactions on Automatic Control* 19.6 (1974), pp. 716–723. DOI: 10.1109/TAC.1974.1100705.

A Visualization of the network

Previously a visualization was shown where each node was displayed probably following the geographical localization. The problem is that the communities were too close to each other. To solve it, another visualization 7 was made with the spring layout to separate physically the communities. On the other hand, the problem with this visualization is that the edges intersect sometimes, so it would seem that the graph is not planar anymore. Again, the nodes with the highest betweennes centrality were highlighted, reflecting the network vulnerabilities.

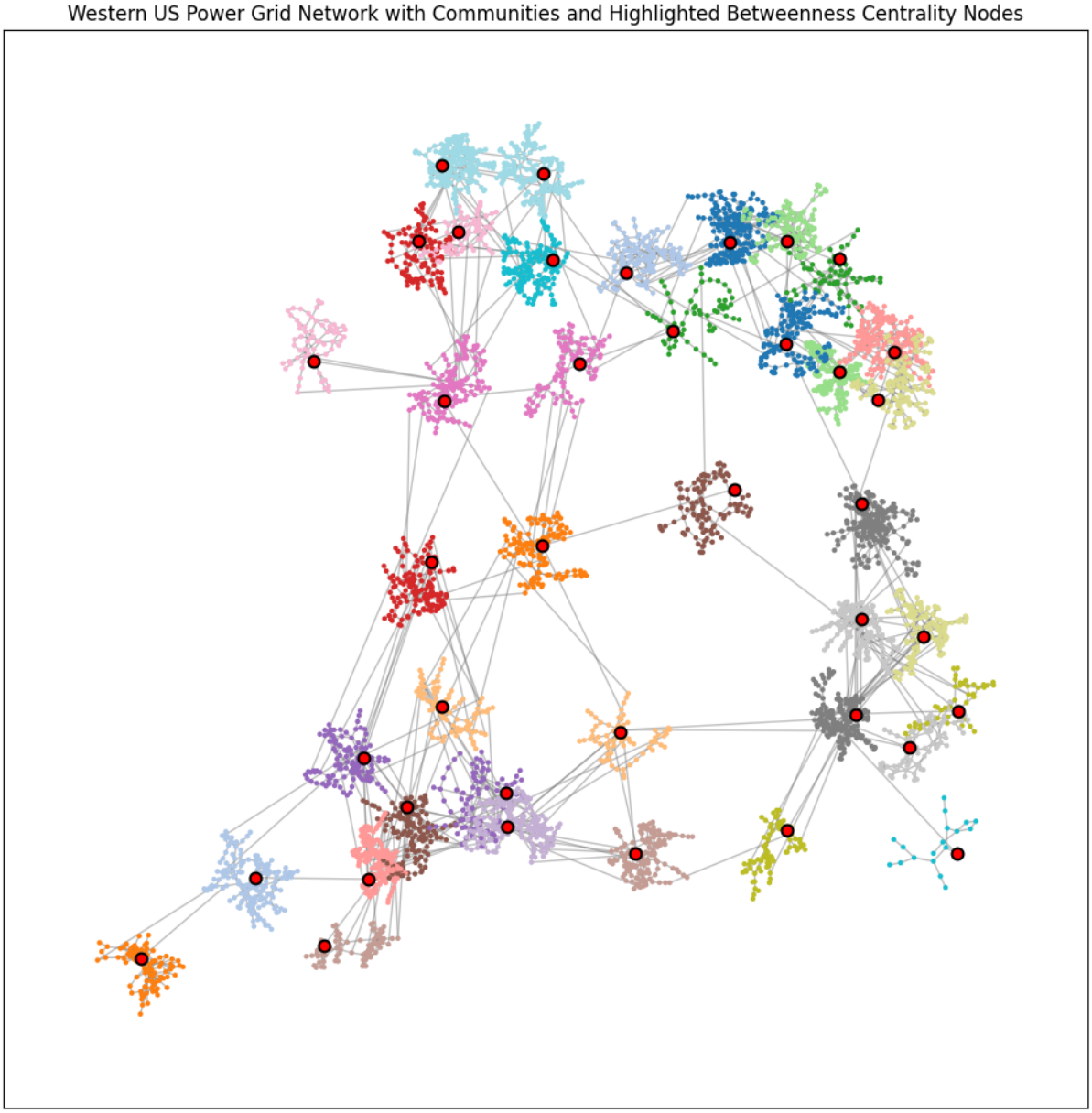

Figure 7: Visualization of the network with the spring layout to separate more the different communites.

Finally, the following visualization 8 is displayed, which is the same from the figure 6 but with added features[6]. In this case, we have various communities represented with different colors and, again, the node with the highest betweenness centrality in each of these is highlighted. As an addition to this, the node with the highest degree centrality has been represented as a hexagon, which is in the top right corner of the figure. In this visualization is also visible the nodes with the highest degree in each

[6]The reason why this visualization is displayed in the appendix and not in the main text is because the extra features presented in the figure could distract from the main focus of the visualization, which is to highlight the communities and the nodes with the highest betweenness centrality in the network, representing these the vulnerabilities of the power grid.

community, represented with squares, which do not coincide with those with the highest betweenness centrality. Another feature represented in the figure are the shortest paths between the highlighted nodes as circles in each community as red edges. As we can see there is a lot of overlap between those short paths, which makes sense taking into account that they connect the nodes with the highest betweenness centrality of the network. Lastly, we computed the average degree centrality for each community and displayed in the figure. We see that there are communities with higher average degree centrality than the one with the highest betweenness centrality node. Of course, they are two different types of centrality so there is nothing wrong with it.

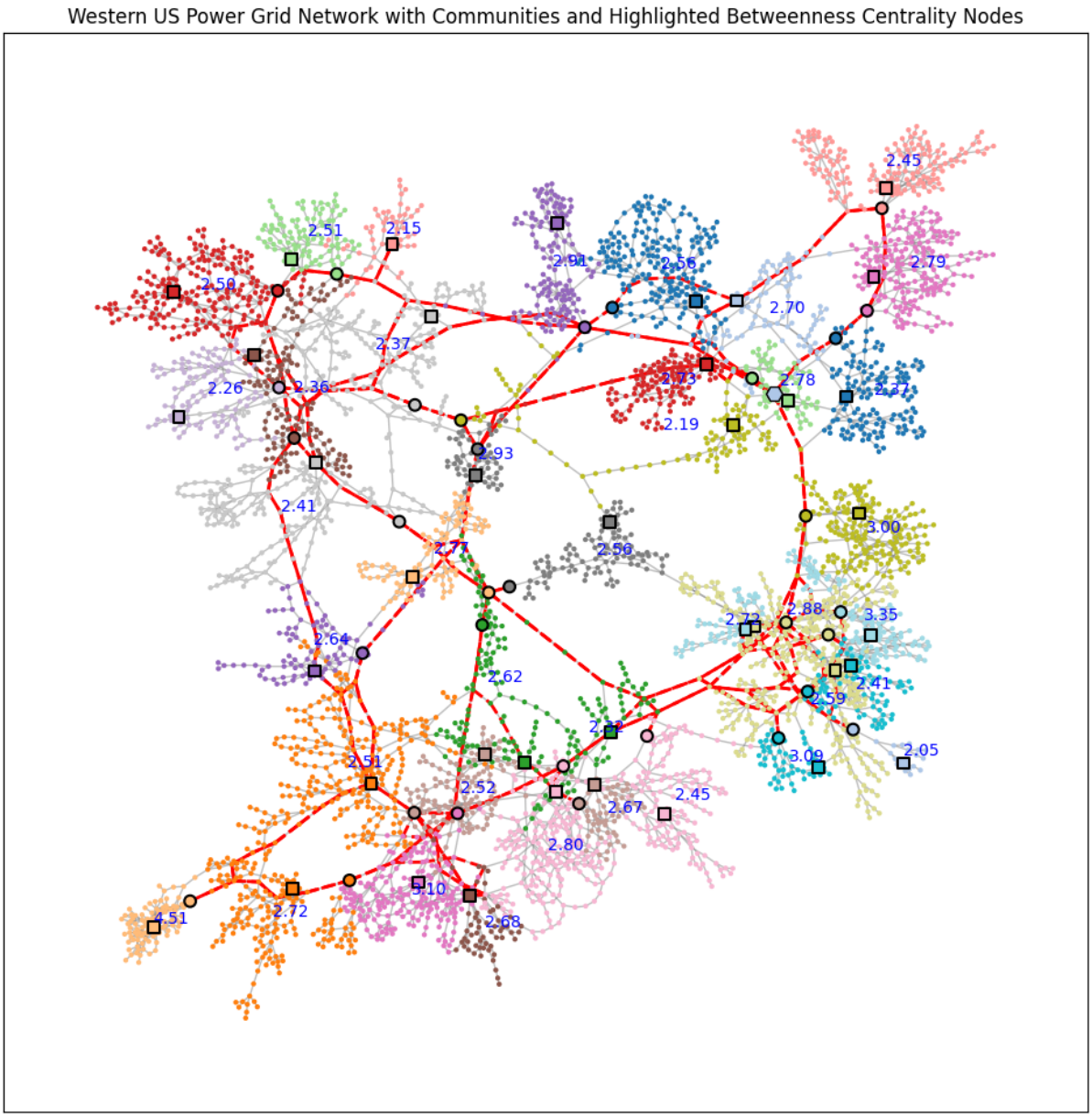

Figure 8: Visualization of the network with the real world layout and with extra features.